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Application of Capacitance Resistance Modeling to Understand Reservoir Connectivity in Large Peninsular Malaysia Brown Field

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Abstract

Capacitance Resistance Modelling (CRM) is a data-driven, signal-processing algorithm grounded in material balance and electrical engineering physics. It is used to evaluate interwell connectivity in waterflood fields, estimating allocation factors from injectors to producers without requiring geological data. CRM is widely adopted in the oil and gas industry due to its low computational cost and minimal input requirements—only production, injection, and pressure data are needed. This study applies CRM in a mature offshore field in Malaysia, with production history exceeding 30 years. The field is dissected by NE-SE faults, creating multiple compartments. The scope of this study includes understanding injector-to-producer support relationships, validating fault transmissibility assumptions, identifying poorly swept or undrained zones, and supporting decisions related to infill well placement and injector optimization.

CRM is based on a simplified material balance within a control volume between injectors and producers. This balance is coupled with a linear productivity equation, resulting in a differential equation solved semi-analytically. The CRM workflow consists of several key steps to ensure robust analysis of interwell connectivity. First, optimal injector-producer pairs are identified within selected cells of the reservoir. This is followed by a thorough quality control process on injection, production, and bottomhole pressure (BHP) data to ensure data reliability. An automated CRM solver is then used to generate and evaluate millions of synthetic scenarios representing a wide range of potential reservoir connectivity patterns.

The best-fit models are selected based on defined history match criteria, including $\pm 10\%$ for reservoir-level production trends, $\pm 15\%$ for well-level trends, and $\pm 20\%$ for connectivity indices. To ensure reliability and forecasting capability, these selected models are further validated through blind tests and compared against available surveillance data, such as tracer results and pressure behavior trends. CRM assumptions and limitations include sufficient signal length, injector rate variability, waterflood pattern stability, and variations in system compressibility. These factors are explicitly reviewed to ensure case-specific adaptation.

The CRM analysis provided valuable insights into fault transmissibility and interwell flow behavior. Fault A was interpreted as non-sealing, as the X3 injector demonstrated strong connectivity with the SX-6 and SX-1 producers, with CRM signals aligning closely with observed production trends. In contrast, Fault

B appeared to act as a sealing barrier, as there was no measurable response from the X6 injector to the SX-10 producer, indicating restricted flow across the fault within the E11 reservoir.

Results from other cells indicated injector inefficiencies, H₂S-related flow path risks, and confirmed justifications for proposed infill wells. For example, low support from X6 to SX10 strengthened the case for additional injection near OPX and OPX2. The CRM application enabled improved understanding of flow patterns, fault behavior, and sweep efficiency. Recommendations included restoring idle injectors, revising waterflood strategy, and refining geological models based on inferred connectivity.

This case study demonstrates a novel application of CRM in a Southeast Asian offshore field characterized by complex faulting. The CRM analysis proved effective in assessing fault transmissibility, offering valuable insights into reservoir compartmentalization. Additionally, CRM-derived connectivity maps provide a strategic foundation for understanding and optimizing the waterflood system. By leveraging historical field data, CRM helps reduce geological uncertainty and supports development planning ahead of detailed field model design.

The work supports CRM's role as a fast, scalable tool for reservoir diagnostics and a valuable complement to reservoir modeling workflows.

Introduction

According to (Gözel 2015), waterflooding remains one of the most widely adopted secondary recovery methods globally. Various approaches, models, and frameworks have been developed to evaluate and forecast the performance of waterflooded reservoirs, aiming to optimize production strategies. One such method is streamline simulation, as illustrated in Fig. 1.

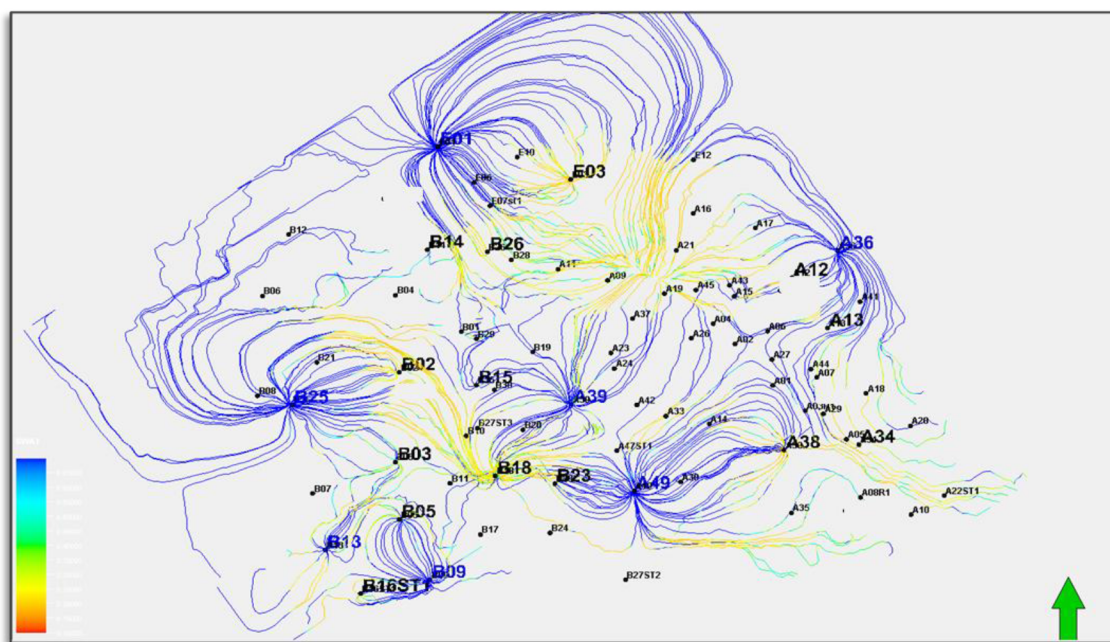


Figure 1—Streamline simulation of Field D Reservoir (BFA Study, 2015)

Traditional waterflood reservoir simulations rely on integrated analyses from petrophysical, geophysical, and geological data. These simulations require extensive input data—such as relative permeability, PVT, and detailed rock and fluid properties and are often time-intensive due to their complex workflows.

As an alternative, the Capacitance-Resistance Model (CRM) offers a data-driven, quantitative approach to assess inter-well connectivity. Based on signal-processing principles, CRM treats injection rates as input signals and production rates as output responses. It enables rapid simulation across multiple scenarios to

evaluate flow patterns and connectivity between injectors and producers. The insights gained can support decisions on adjusting injection or production rates, identifying candidates for re-entry or sidetracking, and optimizing the placement of new injectors or producers.

This paper presents a case study on the application of CRM in a water-injected field located offshore Peninsular Malaysia, in approximately 76 meters of water depth in the South China Sea. First oil was achieved over 30 years ago. The field spans an area of approximately 30 km in length and 5 km in width, characterized by an East-West trending anticline. It is intersected by numerous NE-SE trending faults, which subdivide the field into several fault blocks.

A major fault, with a large throw, separates Fault Block 1 (FB1) from Fault Block 2 (FB2), as illustrated in the Fig. 2. Additionally, the field contains several minor faults. The reservoir was developed through a peripheral water injection strategy initiated in early 1997.

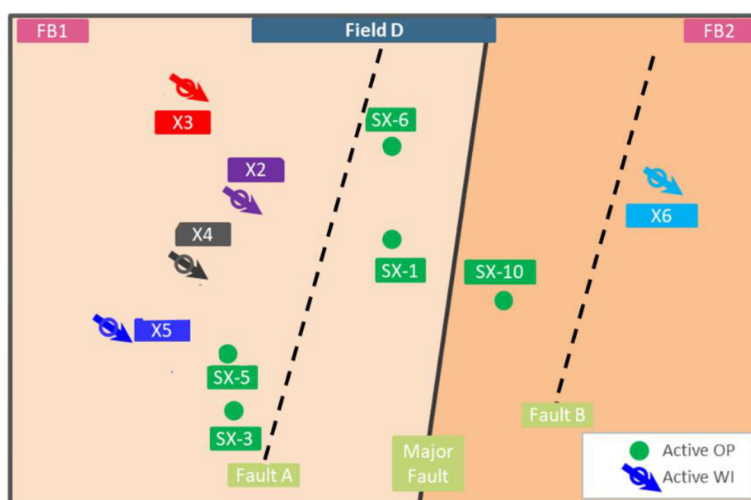


Figure 2—Field D illustration

Capacitance-Resistance Model (CRM)

The Capacitance Resistance Modeling (CRM) algorithm is grounded in material balance principles and electrical engineering concepts, particularly signal-processing techniques. It simulates millions of iterations involving reservoir, injector, and producer parameters to capture the response signals generated by injectors in the corresponding producers.

As a data-driven, quantitative method for assessing inter-well connectivity, CRM serves as an alternative to conventional reservoir simulation. By treating injection rates as input signals and production rates as output responses, CRM enables fast simulation across multiple scenarios. It provides valuable insight into reservoir flow dynamics and injector-producer relationships, supporting decisions such as optimizing injection and production rates, identifying re-entry or sidetrack opportunities, and selecting optimal locations for new wells.

Objectives

The primary objective is to assess the connectivity between injectors and producers using production, injection, and pressure data only, in which independent of geological assumptions. This approach enables asset and study teams to gain valuable insights into reservoir behavior, including fault transmissibility, compartmentalization, and potential leakages, by analyzing well events and historical trends.

The outcome of this analysis provides:

1. A clearer understanding of injector and producer performance patterns

- Recommendations for production rate optimization, new wells, re-entries, side-tracks, and optimal drilling locations for injectors and producers
- Identification of gaps and opportunities to enhance water injection surveillance programs, particularly by evaluating injection pattern efficiency and individual well injectivity performance

Three key areas (or "cells") within the field were selected for CRM analysis to evaluate inter-well connectivity and enhance understanding of reservoir behavior as seen in Fig. 3. The objectives for each cell are summarized in Table 1.

Table 1—CRM objectives set in the Field (examples)

Cell	Justification	Derisking Scope
Cell 1	- Assess whether well SX-6 receives injection support from X3. - If no support is observed, identify where the injected volume from X3 is being distributed. - Investigate why SX-6 and SX-1 exhibit low and stabilized pressures despite ongoing injection from X3. - Evaluate fault transmissibility between the wells.	Existing WI optimization
Cell 2	- Evaluate whether X6 provides effective support to SX-10. - If minimal support is observed, the findings could justify a water injection (WI) infill to enhance support for proposed producers OPX and OPX2. - Support the potential case for the two proposed infill wells (OPX & OPX2).	Existing WI optimization. Check the possibility of leakages
Cell 3	- Assess whether X3, X4, and X5 are providing effective support to SX-3 and SX-5. - Investigate potential H₂S issues linked to X4, and evaluate the feasibility of replacing X4 with X3 as an alternative injector.	Existing WI optimization and H ₂ S mapping

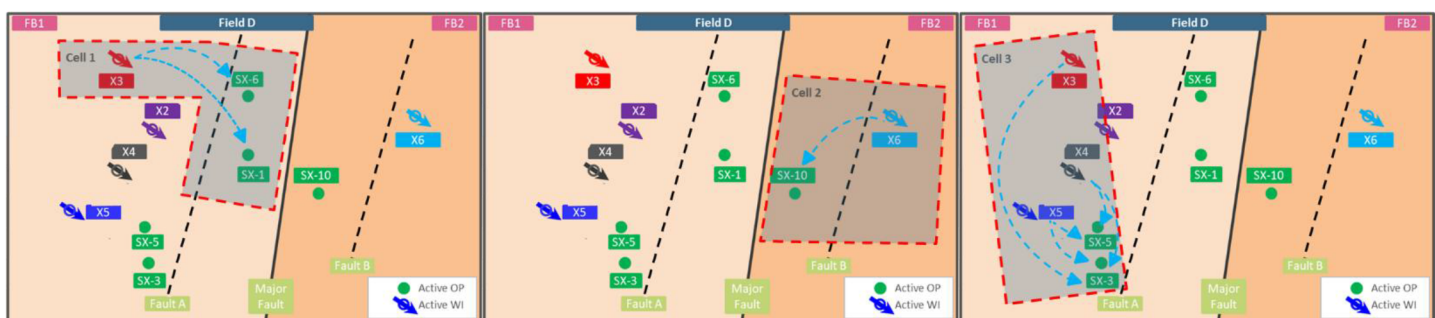


Figure 3—Cell 1, Cell 2 and Cell 3 illustration

Methodology and Limitations

The classical Capacitance-Resistance Model (CRM) is built upon three key components: historical production data, historical injection data, and variations in bottom-hole pressure (BHP) (Sayarpour, Kabir and Lake 2008; Sayarpour, Zuluaga, et al. 2009; Liang 2009; Izgec and Kabir 2010; Holanda, et al. 2018)

The core principle of CRM analysis is to minimize the percentage difference between calculated and actual historical production rates by fine-tuning fitting parameters. The primary parameter is the fraction of

the injection rate that contributes to the production rate at each producer. Other key parameters include the time constant and the productivity index, as depicted in Fig. 4.

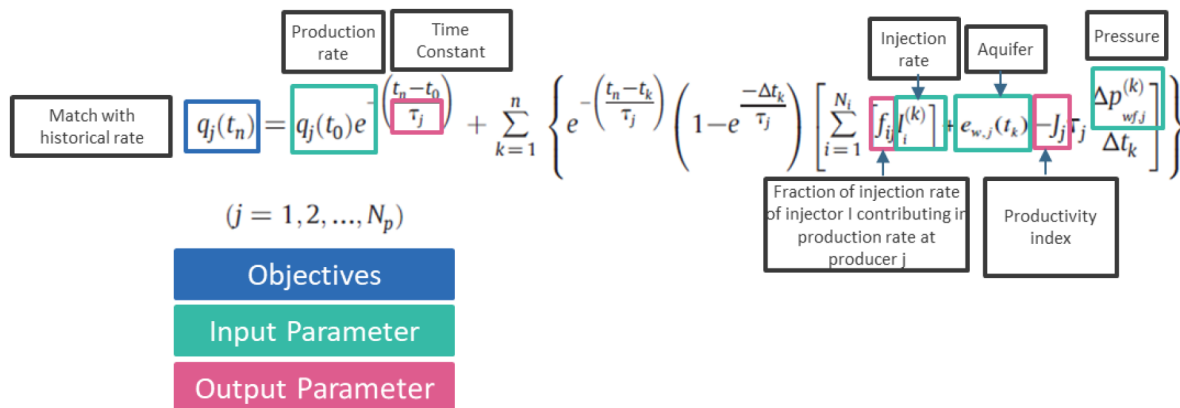


Figure 4—CRM algorithm and calculation (input, output)

When applied to real field data, CRM can simulate millions of scenarios to determine the most accurate combination of reservoir parameters and well-to-well connectivity. This approach enables the model to closely match actual production and injection trends observed in the field. Fig. 5 presents a simplified view of the CRM concept through three example scenarios of inter-well connectivity and reservoir behavior.

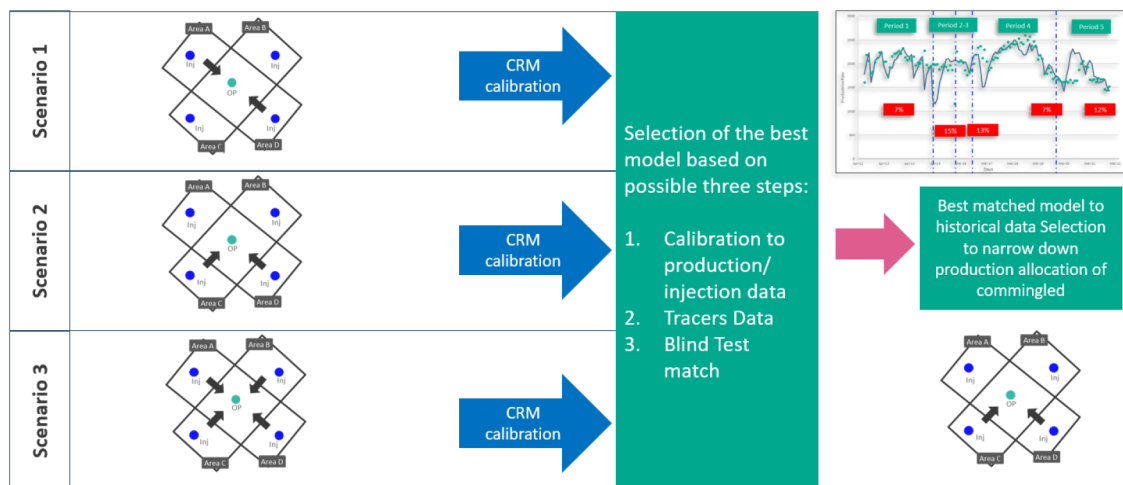


Figure 5—Capacitance – Resistance Model (CRM) concept

The first step in a CRM study is to define the study objectives and understand the reservoir issues. This is followed by data collection, review, and quality control (QC). Once the data is validated, multiple simulation scenarios are created and analyzed using CRM, followed by calibration against historical data. Each scenario undergoes a matching process based on predefined criteria. The final step is the selection of the best-fit model, which calibrated to production and injection data, supported by tracer data validation and blind test performance.

Similar to other history matching techniques, CRM solutions are non-unique—multiple parameter combinations may yield comparable results. Therefore, the quality of CRM results is evaluated using the following key criteria:

Production-Injection Matching

CRM output (production rates) must match historical data within an acceptable tolerance as summarized in Table 2. Note that bottomhole pressure (BHP) is not a default CRM output and is only used as a secondary or supporting input.

Table 2—Success measure tolerance range

History Match Parameter	Tolerance Range
Reservoir level production & injection trend match	$\pm 10\%$
Well level production & injection trend match	$\pm 15\%$
Well pair connectivity index*	$\pm 20\%$

Tracer Data Validation

Tracer test results are used as an independent check to verify injector-producer connectivity predicted by CRM. Streamline simulation results are not recommended for model validation, as they depend heavily on subsurface assumptions and may introduce bias.

Blind Test Validation

A blind test is performed to validate the predictive capability of the CRM model. Typically, 70% of the historical data is used for model training, while the remaining 30% is reserved for blind testing. The training dataset must adequately cover all relevant injector history and their influence on producers. Therefore, for certain producers, the training period may need to be extended to ensure the historical impact of all neighboring injectors is fully captured.

CRM has been successfully applied in numerous oil fields worldwide, with its inter-well connectivity results showing strong consistency with tracer tests and streamline-derived allocation factors. However, successful implementation of CRM on actual field data requires a sound understanding of the model's assumptions and limitations, allowing the workflow to be tailored to specific reservoir conditions. Several key considerations should be taken into account when applying the algorithm:

1. Data availability – Sufficient injection history is essential to ensure that signals from injectors can be detected at the producers, especially for reservoirs with longer response times.
2. Injection rate variability – Greater amplitude and variability in injection rates improve the signal-to-noise ratio, enhancing the model's ability to identify injector-producer relationships.
3. Waterflood pattern changes – Modifications to the injection-production configuration, such as adding new injectors or producers, can impact CRM calibration and must be accounted for.
4. Reservoir compressibility changes – Variations in total system compressibility, often due to gas cap expansion or pressure decline below bubble point, can influence CRM behavior and should be evaluated.
5. Well events and productivity fluctuations – Changes in well conditions, including shut-ins, stimulations, or equipment failures, may affect signal interpretation and need to be integrated into the analysis

Therefore, these highlighted factors must be carefully considered to understand their influence on model parameters, the history matching process, and the customization of the workflow for evaluating inter-well connectivity in the specific reservoir setting. The conceptual workflow implemented in the CRM software consists of the following steps as shown in Fig. 6.

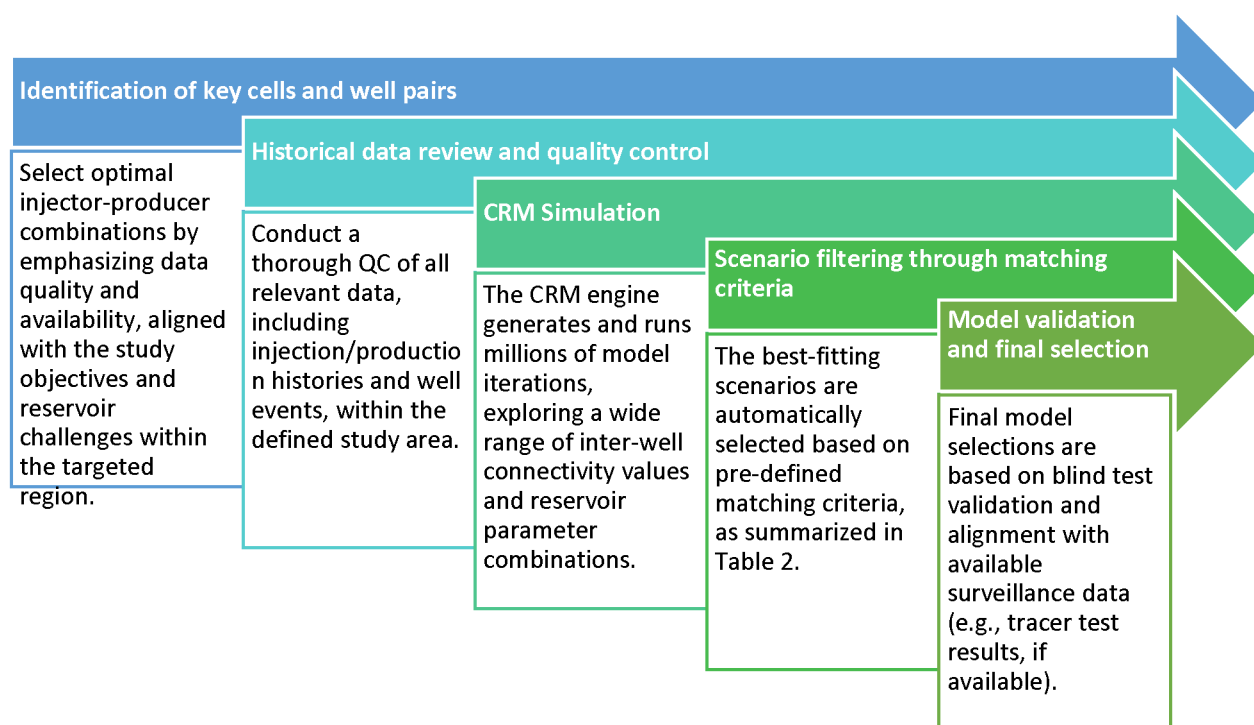


Figure 6—CRM workflow

Result and Discussion

Cell 1

Cell 1 comprises SX-6 as the main producer and X3 as the injector, with a minor fault (Fault A) located between the two wells. The primary objective of this analysis is to assess the transmissibility of Fault A and validate whether it acts as a barrier to fluid flow.

Based on CRM results, X3 demonstrates a clear connectivity with SX-6, where the injection signal aligns well with the production response. This indicates that injection from X3 is effectively supporting SX-6. Therefore, it can be inferred that Fault A is not sealing and allows fluid communication between the injector and producer.

Cell 2

Cell 2 consists of a one-to-one water injection–oil production pair: X6 (injector) and SX-10 (producer), located in the eastern region of Field D. A fault separates the two wells. The primary objective of the CRM analysis is to determine whether X6 provides effective support to SX-10, thereby evaluating the connectivity between the wells and the sealing nature of the fault. This insight is critical for justifying a potential water injection (WI) infill well to support oil production, particularly if SX-10 is currently receiving insufficient support from X6.

Based on the CRM results, no connectivity is observed between X6 and SX-10. The injection signal from X6 does not correlate with the historical production response of SX-10. This suggests that the fault may be acting as a barrier to flow between the two wells.

Cell 3

In Cell 3, the objective is to evaluate the connectivity of injectors X3, X4, and X5 with producers SX-5 and SX-3. This assessment is critical due to a reported H₂S issue that is suspected to be caused by injector X4. As a mitigation measure, X3 is being considered as a replacement for X4. Therefore, understanding the level of connectivity between these wells is essential to guide injection strategy and mitigate H₂S concerns.

Based on the CRM analysis, there is clear connectivity between injector X5 and producers SX-5 and SX-3. The injection signals from X5 show a strong match with the production response from both producers. In contrast, nearby injectors X3 and X4 show minimal to no connectivity, suggesting they contribute little to the production response in this area.

An optimisation study was also carried out to evaluate the impact of injectors X5, X4, and X3 on the production rates of SX-5 and SX-3. Several injection rate scenarios were simulated to understand how changes in injection affect future production performance. The results of this optimisation indicate that increasing injection from X5 leads to higher production rates, confirming its strong support role. Meanwhile, the impact from X4 is minimal, supporting the idea that it could be replaced by X3 without adversely affecting the performance of SX-5 and SX-3.

However, the oil incremental was estimated using limited data and assumptions, such as pressure trends and connectivity coefficients. CRM can be applied to predict oil production profiles (whether increasing or declining) based on injector behavior. It is advisable to conduct a detailed study to obtain a more reliable prediction and accurate estimates.

An example of the final CRM match for one of the producers is shown in Fig. 7. Additionally, Fig. 7 includes fault transmissibility results, providing further insight into reservoir connectivity in this area

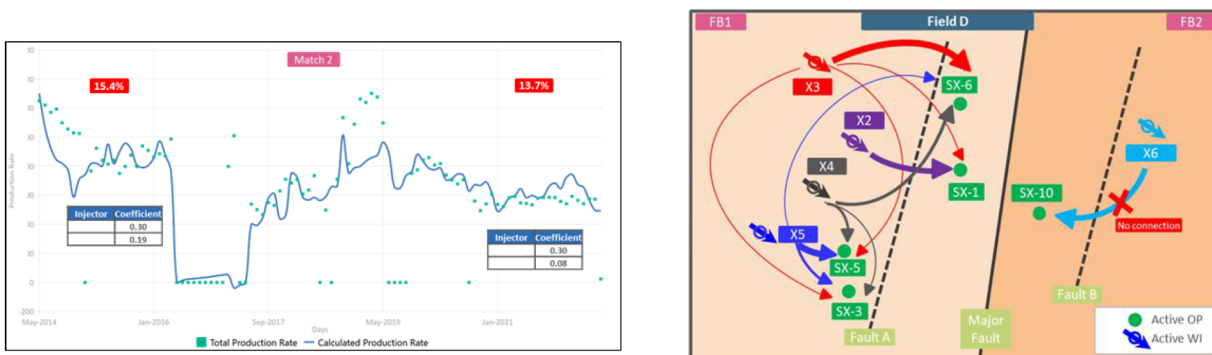


Figure 7—CRM cell example match (model – line vs history (points)) (left) and field map

Conclusions and Recommendations

The application of Capacitance-Resistance Model (CRM) in this field has provided valuable insights into interwell connectivity, fault transmissibility, and the effectiveness of the current waterflood pattern. By leveraging production, injection, and pressure data, independent of geological assumptions, CRM has enabled a detailed evaluation of dynamic reservoir behavior across multiple areas of the field.

Three key cells were analyzed as case studies:

1. Cell 1 demonstrated strong connectivity between injector X3 and producer SX-6, despite the presence of a minor fault. The matched injection and production signals suggest the fault is non-sealing.
2. Cell 2 revealed no observable support from injector X6 to producer SX-10, indicating that the fault between them may be sealing. This result supports the justification for a water injection infill well to improve sweep efficiency in this area.
3. Cell 3 highlighted significant contribution from injector X5 to producers SX-5 and SX-3, with X3 and X4 providing minimal support. Given the potential H₂S issue associated with X4, CRM results support replacing X4 with X3, as it would not adversely affect production performance. Optimization scenarios confirmed that increasing X5 injection is the most effective strategy to boost production in this cell.

Overall, CRM has proven to be a powerful tool for diagnosing injector-producer relationships, guiding infill well decisions, validating fault behavior, and refining waterflood management strategies. In addition, it serves as a fast simulation tool, enabling rapid analysis and decision-making without the need for complex geological models. These insights support the design of a more efficient reservoir surveillance and development plan, ultimately enhancing field recovery and optimizing resource management.

The implementation of new production data analysis techniques, such as CRM analysis, to analyze fields with long water injection history enabled the team to gain a deeper understanding of reservoir flow dynamics and optimize decisions on further fields development (restore an idle water injector, new infill wells location, wells efficiency, etc.).

To enhance reservoir understanding and improve management of Field D, several actions are proposed. First, it is recommended to extend CRM analysis to additional cells, particularly those involving oil producers in the northern region of the field. This will improve the overall understanding of reservoir behavior and inter-well connectivity. In parallel, performing multi-well pressure deconvolution is suggested to extract deeper reservoir insights, further supporting reservoir characterization and monitoring efforts.

In terms of data acquisition and well intervention, pulse testing between selected oil producers and water injectors is advised. This will provide valuable information on fault block transmissibility and confirm inter-well communication pathways. An integrated subsurface study is also recommended, where the existing geological, static, and dynamic models should be merged with CRM results, particularly the insights on connectivity and fault behavior. With this enhanced model, oil production optimization can be performed more reliably, considering both well-level constraints and surface facility limitations, either through streamline simulation or full-field dynamic models. These actions collectively aim to validate remaining resources in the field and support more informed development planning.

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